Method of expressing concentration of solution

81. (d) Molality $(m) = \frac{w \times 1000}{mW} = 14.05.$

82. (b) $N_1V_1 = N_2V_2$

$10 \times 10 = 0.1(10 + V)$

$V = \frac{10 \times 10}{0.1} - 10 = 1000 - 10 = 990ml.$

83. (b) Sum of mole fraction is always 1.

84. (b) An increases in temperature increase the volume of the solution and thus decreases its molarity.

85. (d) 10^3 parts of $CaCO_3$ has number of parts = 10

10^6parts of $CaCO_3$ has number of parts

$= \frac{10}{10^3} \times 10^6 = 10{,}000ppm.$

86. (d) $X = \frac{n}{n+N}$

$n = \frac{w}{m} = \frac{3.65}{36.5} = 0.1, \quad N = \frac{W}{M} = \frac{16.2}{18} = 0.9$

$X = \frac{0.1}{0.1+0.9} = 0.1.$

87. (d) 10% glucose solution means $10g = \frac{10}{180} mole$ in 100 cc. *i.e.*, 0.1 *litre*

Hence 1 *mole* will be present in $\frac{0.1 \times 180}{10}$=1.8 *litre*.

88. (e) For methyl alcohol $N = M$.

89. (b) Mole fraction of glucose = $\frac{n}{n+N}$

$= \frac{0.01}{0.01+5} = 0.00199$

90. (b) Mole of urea $= \frac{6.02\times10^{20}}{6.02\times10^{23}} = 10^{-3}$ moles

Conc. of solution (in molarity) $= \frac{10^{-3}}{100} \times 1000 = 0.01M.$

91. (a) Gram molecule of SO_2Cl_2 = 135

$n = \frac{w}{m} = \frac{13.5}{135} = 0.1.$

92. (d) 1000 ml of 1 N oxalic solution = 63 g

500 ml of 0.2 N oxalic acid solution

$= \frac{63}{1000} \times 500 \times 0.2 = 6.3g.$

93. (a) Mole fraction at $C_6H_6 = \frac{\frac{7.8}{78}}{\frac{7.8}{78}+\frac{46}{92}} = \frac{1}{6}.$

94. (c) $X_{H_2O} = \frac{n_{H_2O}}{n_{H_2O}+n_{C_2H_5OH}+n_{CH_3COOH}}$

95. (a) $M_1V_1 = M_2V_2$

$i.e.\ 5 \times 1 = M_2 \times 10 \Rightarrow M_2 = 0.5$

Normality of the solution $= \frac{0.5}{2} = 0.25.$

96. (a) $M = \frac{w\times1000}{m\times\text{Volume in}ml.} = \frac{1\times1000}{40\times250} = 0.1M.$

98. (d) $N = \frac{w \times 1000}{eq.wt. \times \text{volume in} ml.} = 0.33N.$

99. (b) Mole of $HCl = \frac{1.2046 \times 10^{24}}{6.023 \times 10^{23}} = 2mole$

Normality = molarity×basidity or acicity $= 2 \times 1 = 2N$

100. (d) $10N$ = Deca-normal, $\frac{1}{10}N$ = Deci-normal.

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